Final Project

A Quantitative Analysis of Air Quality in Harrisonburg

Timothy Tarter, Sam DiBiase

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James Madison University

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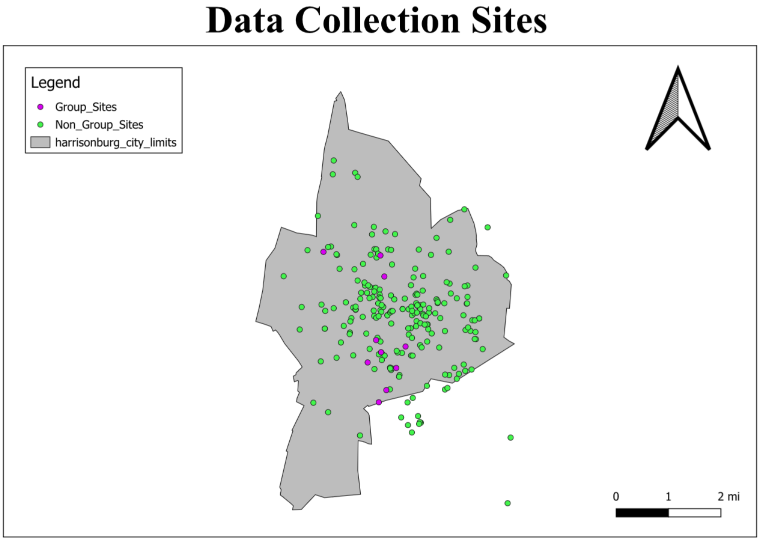
Our Measurements



Our Measurements

Tim's Measurements:

- Taken late at night
- Taken on a humid and stormy day

Sam's Measurements:

- Taken in the early afternoon
- Taken on a sunny and warm day

PM 2.5 Analytics

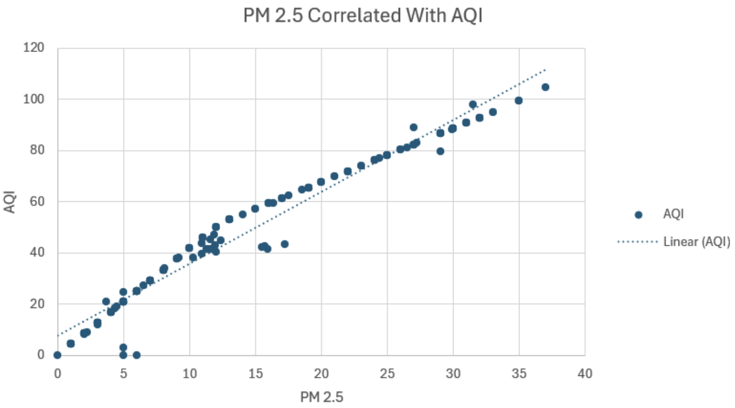
True PM 2.5	Predicted PM 2.5	Absolute Error
5	26.01	21.01
37	19.71	17.29
35	15.55	19.45
32	14.27	17.73
6	14.20	8.20
33	14.19	18.81
7	12.77	5.77
7	12.31	5.31
33	11.69	21.31
5	8.14	3.14

PM 2.5 Analytics

- Median absolute error value of 13.8 micrograms per cubic meter.
- Large differences due to geographic discrepancies in collections.

Other Interesting Results

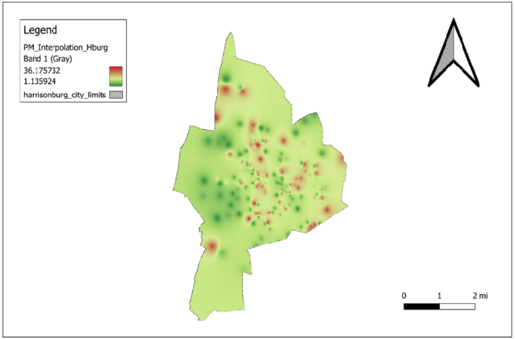
- Correlation between PM 2.5 and AQI: pseudo-logarithmic.



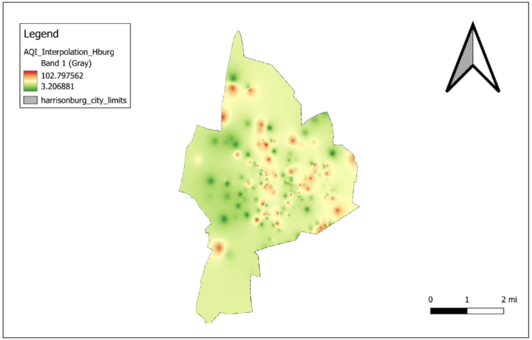
Other Interesting Results

- Correlation between PM 2.5 and AQI: geographically related.

PM 2.5 Interpolated Heatmap

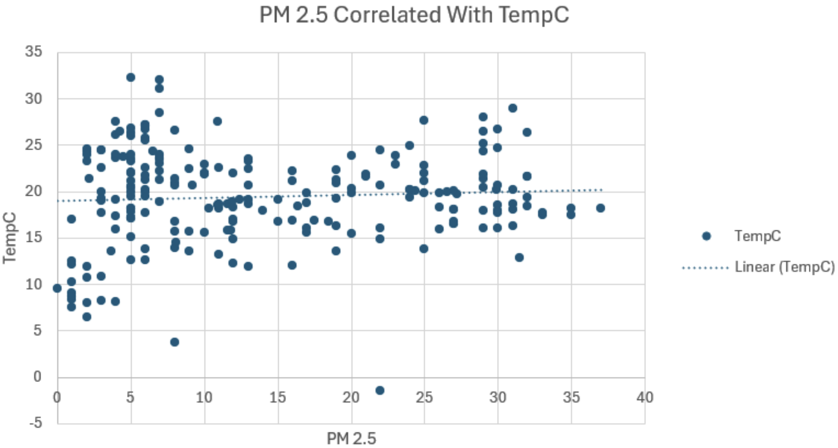


AQI Interpolation Heatmap



Other Interesting Results

- Temperature: completely uncorrelated with PM 2.5.



Final Interesting Results...

Statistic	PM 10	PM 2.5	Ozone	AQI	Temperature (Celsius)
Median	13	11	0.2	42.2	19.8
Maximum	45	37	36.2	104.7	32.2
Minimum	0	0	0	0	-1.5

What if Ozone Hit 36.2 PPM?

AI Overview

If ground-level ozone concentrations reached **37 ppm (parts per million)**, it would be a catastrophic environmental and health catastrophe, as this level is **hundreds of times higher** than current, already dangerous, air quality standards.

For context, the EPA considers ozone levels over **0.086 ppm** (86 ppb) as "unhealthy," and immediate, acute, and likely lethal respiratory effects are documented in humans at levels far lower than 37 ppm. U.S. Environmental Protection Agency (.... +2

Here is what would happen if ozone concentrations were 37 ppm:

1. Instant and Catastrophic Human Health Impact

- **Acute Lethality:** Exposure to ozone levels in the range of 5–50 ppm is known to be immediately dangerous to life and health (IDLH). At 37 ppm, exposure for even a short time would likely cause fatal pulmonary edema (fluid in the lungs) and severe acute respiratory distress.
- **Respiratory Destruction:** Ozone is a powerful oxidant that "burns" lung tissue. At 37 ppm, people would experience severe chest pain, extreme shortness of breath, coughing, and throat damage instantly upon breathing the air.
- **Severe Toxicity:** Breathing would cause immediate damage to airways, causing the muscles to constrict and trapping air in the lungs. U.S. Environmental Protection Agency (.... +3

References

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